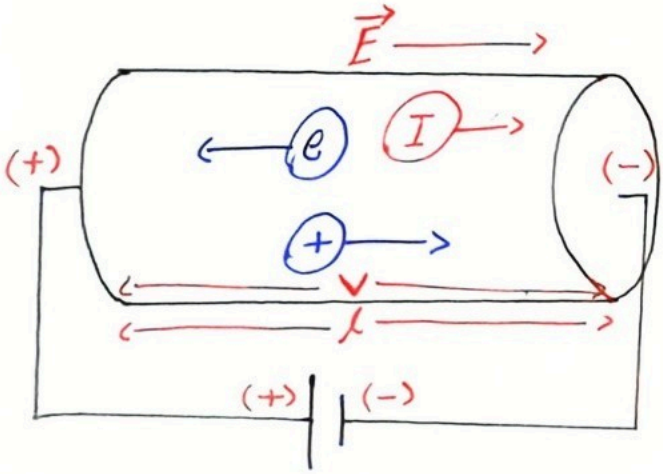
Current electricity

⟹ Basics

⟹ How do electric fields occur?



e - electrons
+ - positive charges
$\vec{E}$ - electric field
I - eletric current
V - Potential difference
l - length of the conductor

⟹ Equation

$$\boxed{\vec{E} = \frac{V}{l}}$$

⟹ Eletric Current

$$\boxed{\begin{aligned} Q &= It \\ I &= \frac{Q}{t} \end{aligned}}$$

Q - Amount of charge
I - eletric current
t - time

⟹ Units

Q - C
I - Cs^{-1} / A

Cs^{-1} - Coulomb per second
A - Ampere

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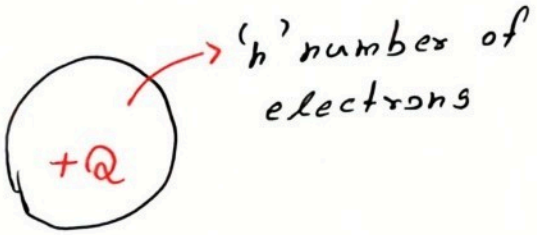
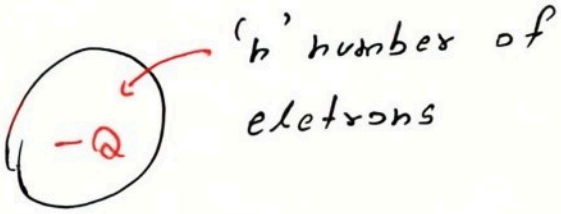
⇒ Electric Charge (Q)

→ When eletrons are removed.



$$Q = ne$$

→ When eletrons are added.



$$Q = ne$$

⇒ Derivations ($Q = It$ / $Q = ne$)

$$It = ne$$

$$I = \frac{ne}{t}$$

$$t = \frac{ne}{I}$$

$$e = \frac{It}{n}$$

$$n = \frac{It}{e}$$

You don't need to memorize the derivations, you just need to apply them based on what the question ask

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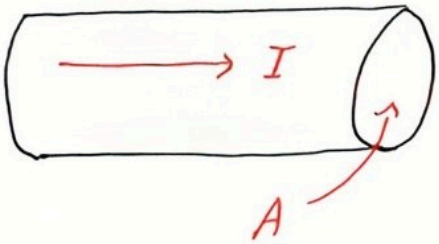
⇒ Current Density (J)



$$J = \frac{I}{A}$$

Similar to mass density (ρ)

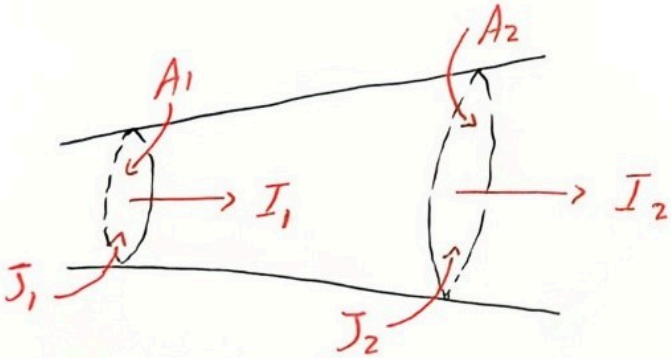
$$\rho = \frac{m}{V}$$

→ Units,

$$J = Am^{-2}$$

Am^{-2} - Amperes per square meter

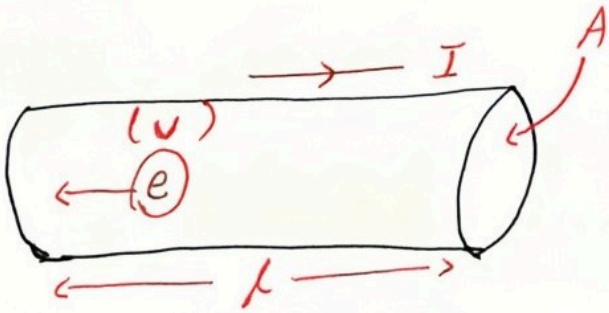
⇒ If the conductor is continuous, 'I' stays Constant



$$I_2 = I_1$$

$$J_2 A_2 = J_1 A_1$$

⇒ Relationship between mean drift velocity and eletric current



$$I = neAv$$

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→ Relationship between drift velocity and current density,

$$I = neAv$$

$$\frac{I}{A} = nev$$

$$J = nev$$

⇒ Root mean square Speed (V_{rms})

→ Mean kinetic energy,

$$= \frac{3}{2} kT$$

k - Boltzmann's constant

→ Equation,

$$\frac{1}{2} m_e V_{rms}^2 = \frac{3}{2} kT$$

$$V_{rms}^2 = \frac{3kT}{m_e}$$

$$V_{rms} = \sqrt{\frac{3kT}{m_e}}$$

m_e - mass of a electron

T - Absolute temperature of conductor

→ Boltzmann's Constant,

$$k = \frac{R}{N_A}$$

R - Universal Gas Constant

N_A - Avogadro's Constant

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→ Basics,

$$n = \frac{m}{M}$$

n - moles

m - mass of substance

M - molar mass of the substance

$$N = n \times N_A$$

N - Number of molecules or atoms

N_A - Avogadro's constant

⇒ Electric Resistance (R)

$$R = \rho \frac{l}{A}$$

Unit → Ω (ohm)

→ electric resistivity,

$$\rho = \frac{RA}{l}$$

Unit → Ωm (ohm meter)

→ electric conductance,

$$G = \frac{1}{R}$$

Unit → Ω^{-1} / S (- per ohm) (- Siemens)

→ electric conductivity,

$$\sigma = \frac{1}{\rho}$$

Unit → $\Omega^{-1} m^{-1}$ / $S m^{-1}$

(- per ohm per meter)

(- Siemens per meter)

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⇒ Ohm's Law

$$V = IR$$

⇒ Variation of resistance with temperature

$$R_\theta = R_0 (1 + \alpha \theta)$$

R_θ - Resistance at temperature θ

R_0 - Resistance at $0°C$

α - The temperature coefficient of resistance

θ - The temperature

Units - "α" - $°C^{-1}$ / K^{-1}

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